

Fig. 3: Effect of temperature on the state of matter

combustion products including the polluting flue gas. The molecular dissociation zone starts above 2700°C. Any temperature below this will produce incomplete dissociation. Thus with the temperatures achieved in a PGV system (above 2700°C), all the molecules are totally dissociated. Fig. 3 shows the effect of temperature on the

state of matter as the temperature rises to about 10,000°C

General classification and treatment of wastes in plasma gasifier

All wastes may be generally classified into organic or inorganic wastes. Organic wastes are associated with living things, i.e., having biological origin, e.g., flora, fauna, food and living beings. These all have carbon-based compounds.

On the other hand, inorganic wastes are those having non-biological (or geological) origin, e.g. minerals, rock, sea and air. Most inorganic

compounds lack carbon, e.g., salt (NaCl) and ammonia (NH₃); however, a few, such as carbon dioxide (CO₂), do contain it, but it is never attached to hydrogen atoms as in hydrocarbons. Inorganic molecules tend to have a relatively small number of atoms as compared to organic molecules.

The plasma torches that operate at very high temperatures (between 5000°C and 100,000°C) can process both inorganic and organic wastes including industrial and nuclear wastes at atmospheric pressure. Transformation of organic and inorganic matter in the plasma reactor results in the following:

1. The extreme temperatures generated using a plasma torch system (reactor), transform the

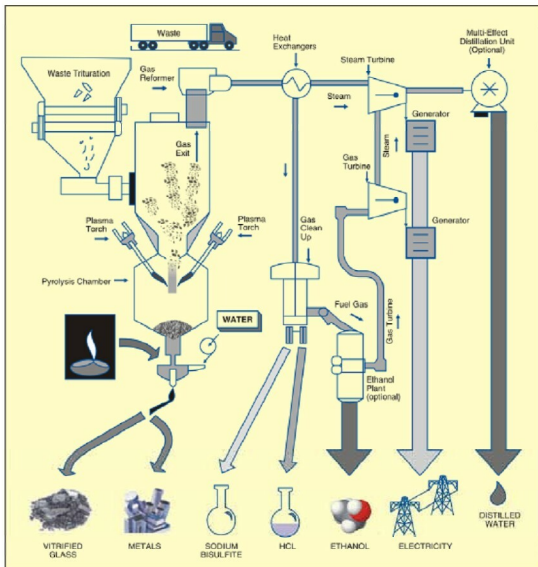


Fig. 4: Artistic view of plasma trash processing system including generation of byproducts